

01 The Mental Model

Git = a tool that tracks changes to files. Like Microsoft Word's "Track Changes," but for whole projects.

GitHub = a website that hosts Git projects in the cloud. Like OneDrive — but for code and notebooks.

GitHub.com → Your Studio → Open notebook

02 The Five Commands You Need

`git clone <url>`

Download a project from GitHub into your Studio. You'll use this most.

`git pull`

Refresh a project you already cloned to get the latest version.

`git status`

Show what's changed. Red text here is information, not an error.

`ls`

List files in the current folder. (This is a terminal command, not Git.)

`cd <folder>`

Change directory — move into a folder. Use `cd ..` to go up one level.

03 Vocabulary, Decoded

Repository (repo) A project folder tracked by Git.

Clone Make your own working copy of a repo on your machine.

Pull Bring down the latest changes from GitHub.

Commit A saved snapshot of your changes. (*Advanced — session two.*)

Push Upload your saved snapshots back to GitHub. (*Advanced.*)

04 Recipe: Get the AWS Tutorials

CLONING THE SAGEMAKER EXAMPLES

- 1 In SageMaker Studio, open the **File** menu → **New** → **Terminal**.
- 2 Type or paste this command and press Enter:


```
git clone https://github.com/aws/amazon-sagemaker-examples.git
```
- 3 Wait for the download to finish (about 30 - 60 seconds).
- 4 Run `ls` — you should see a folder named `amazon-sagemaker-examples`.
- 5 Run `cd amazon-sagemaker-examples` to enter the folder.
- 6 Open the file browser on the left and double-click any `.ipynb` file to begin.

05 Recipe: Refresh the Tutorials

WHEN AWS UPDATES THE MATERIALS

- 1 Open a Terminal in Studio.
- 2 Run `cd amazon-sagemaker-examples` to enter the folder.
- 3 Run `git pull` to download the latest changes.
- 4 That's it. New and updated notebooks are now available.

THREE THINGS TO REMEMBER

You will not break anything by trying. · Red text in the terminal is usually information, not an error. · If lost, run `pwd` to see where you are.

Stuck? *Start here.*



Match what you see on screen to a card below.

Each card shows the message, what it actually means in plain English, and the one thing to try first. If none of these match, jump to the Universal Fixes at the bottom.

YOU SEE `command not found: git`

Wrong window. You're typing into something that isn't a terminal — likely a notebook cell or a chat box.

FIX Open a real terminal: in Studio, **File → New → Terminal**. Try the command there.

YOU SEE `fatal: destination path '...' already exists and is not an empty directory`

You already cloned this. Git won't overwrite an existing folder of the same name.

FIX Don't clone again. Just enter the folder: `cd amazon-sagemaker-examples`

YOU SEE `fatal: not a git repository`

Wrong folder. You're running a Git command from a folder that isn't a Git project.

FIX Run `ls` to see folders, then `cd` into the project folder before retrying.

YOU SEE `Authentication failed / Permission denied`

You're trying to access a private repo. Public AWS samples should not need a login.

FIX Double-check the URL is exactly the one on the cheat sheet. If still stuck, ask the instructor — Personal Access Tokens come in workshop two.

YOU SEE `No such file or directory`

Typo, or you're not where you think you are. Folder names are case-sensitive.

FIX Run `pwd` to confirm location, then `ls` to see what's actually there. Match capitalization exactly.

YOU SEE `Lots of red text after running git status`

Not an error. Red here just means "files that have changed." Git is showing you the news, not yelling.

FIX Read it like a status report. If you didn't intend to change anything, leave it alone and proceed.

YOU SEE `Could not resolve host: github.com`

Studio lost internet. Either your space lost network access or GitHub is briefly unreachable.

FIX Wait 30 seconds and retry. If it persists, stop and restart your JupyterLab space.

YOU SEE `My notebook isn't showing up in the file browser`

It's there — just one folder deeper. Cloned repos always create a subfolder.

FIX In the file browser on the left, double-click `amazon-sagemaker-examples` to open it.

Universal Fixes (try these first, in order)

- 1** Run `pwd` to see your current folder.
- 2** Run `ls` to see what's in it.
- 3** Re-read the error message slowly — out loud.
- 4** Check capitalization and punctuation in your command.
- 5** Close the terminal, open a fresh one, try again.
- 6** Raise your hand. You are not breaking anything.

*When in doubt, the answer is almost always **cd** or **ls**.*